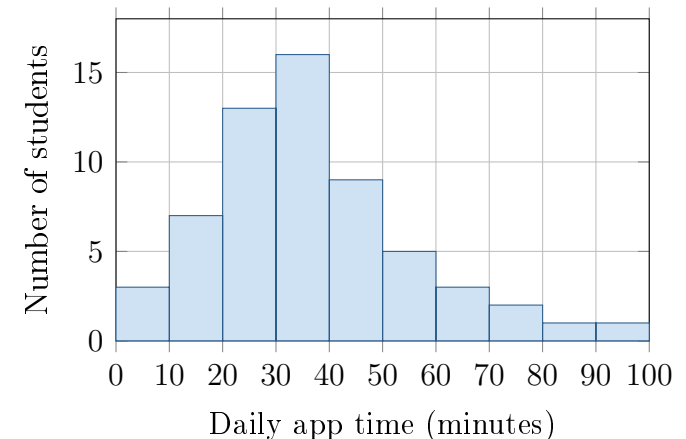


Check the Model

Unit 2 · Topic 2.11 · TODAY'S PROBLEM

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes. Table A gives a left area of 0.9938 for $z = 2.5$.

Ben: “About 16% should be above 52 minutes.” **Ava:** “For 70 minutes, $z = 2.5$, so less than 1% should be above 70.”



FIRST TAKE (5 min, on your sheet)

Do the times look balanced around 40 minutes, or does one side stretch farther? Point to the bars.

- DOK 1** (a) State the proposed model's mean μ and standard deviation σ , with units.
- DOK 2** (b) Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.
- DOK 3** (c) Would you use Ben's and Ava's model results to describe these students? Use the histogram's shape and the observed versus predicted percent above 70. Explain what a reader could get wrong about long app times.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

